



PROFILE

I am an MD candidate in the 8-year program at Fudan University. My research focuses on the intersection of clinical neuroscience and machine learning, specifically the development of multi-modal foundation models and agentic systems for analyzing spatial transcriptomics and neurobiological data. I am driven to apply data-driven methodologies to investigate the pathological mechanisms of brain disorders within the MSRA AI and the Brain group.

CONTACT

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RESEARCH INTERESTS

Multi-modal Representation Learning
Agentic Systems & LLM Reasoning
Spatial Transcriptomics & Bioinformatics
Neurodegenerative Disease Mechanisms

ACADEMIC ADVISORS

Prof. Jintai Yu
Official MD Supervisor
[Institute of Medical Genetics, Fudan University](#)

Prof. Zhiyuan Yuan
Primary Research Mentor
[Fudan Data-Driven Future Lab](#)

Prof. Bin Song
Former Research Supervisor
[Massachusetts General Hospital](#)

Prof. Fei Lan
Early Wet Lab Mentor
[Institutes of Biomedical Sciences](#)

SKILLS

Clinical & Laboratory Skills

Basic Clinical Skills (Physical Exam, etc.)

Basic Laboratory Techniques

AI & Data Analysis

Python, R, C, TypeScript

Agentic Frameworks (LangGraph, ReAct)

Neural Networks (Transformers, GNNs) & Multi-GPU Training

Multi-modal Contrastive Learning

Statistical Analysis

Bioinformatics & Systems

Spatial Transcriptomics Analysis

Digital Pathology & Image Analysis

Remote HPC & Network Optimization (Linux)

Language Proficiency

English: TOEFL 105+

Mandarin / Cantonese / Teochew: Native

Spanish: Basic (A1 level)

Jiahao Ji

EDUCATION

Fudan University 2019.09 – 2027.06
Doctor of Medicine (M.D.), 8-Year Clinical Medicine Program

University of California, Los Angeles (UCLA) 2021.09 – 2022.02
Exchange Program (UCEAP) | Intensive lab-based research and coursework

Guangzhou No.2 High School 2016.09 – 2019.06
High School Diploma

PUBLICATIONS & MANUSCRIPTS

SpatialDataAgent: Autonomous Curation of Spatial Omics Data
Sole Developer | Prepared for *Nature Methods* (2026.04)

Independently developed an autonomous framework using an **Autoresearch-style iterative ReAct loop** to curate spatial transcriptomics datasets. Recovered 769 H&E-ST paired datasets from GEO (+141% over manual methods) and generated HESRT, a datalake containing 29.2 million spots and 29.9 gigapixels of histological data.

Spatiotemporally resolved transcriptome unveils reactive subpallial microglia underlying cholinergic vulnerability in Alzheimer’s disease
Co-first Author | Under Review at *Cell* (Submitted 2026.04)

Combined in situ β -amyloid staining with single-nucleus sequencing to generate the largest spatiotemporally resolved AD transcriptomic atlas to date, integrating **12.36 million spatial profiles**. Established the mechanistic link between microglia microenvironments and basal forebrain cholinergic neuron vulnerability.

SELECTED ENGINEERING PROJECTS

ASSA: Autonomous Self-Sovereign Agent [github.com/Biogod2020/ASSA](#)

Developed a robust agentic framework featuring hierarchical memory systems and skill-generation loops for autonomous adaptation.

Spatial-CLIP: Multi-modal Contrastive Learning [github.com/Biogod2020/Spatial-CLIP](#)

Fine-tuned vision-language encoders to align histopathology images with spatial transcriptomics data, enabling cross-modal retrieval.

RESEARCH & CLINICAL EXPERIENCE

Pr. Jintai Yu & Pr. Zhiyuan Yuan's Lab 2023.09 – Present

Research Project Leader | Focus: Alzheimer's Spatial Intelligence

Operating at the nexus of clinical neurology and computational biology. As the official student of Dr. Jintai Yu, I synergized his extensive Alzheimer's clinical/biomarker insights with the cutting-edge spatial multi-omics technologies guided primarily by Dr. Zhiyuan Yuan.

Spearheaded the computational mapping of the β -amyloid plaque microenvironment. Developed robust algorithms and utilized transformer networks to align high-resolution whole-slide imaging (WSI) with spatial transcriptomic coordinates, uncovering spatiotemporal dynamics and regional neuronal vulnerabilities in AD pathogenesis.

Pr. Bin Song Lab 2021.10 – 2023.06

Undergraduate Researcher | Focus: Parkinson's Disease & scRNA-seq

Engineered scRNA-seq analytical pipelines to identify crucial genetic markers, enhancing the purity, safety, and treatment efficacy of autologous iPSC-derived DA progenitor transplantation in Parkinson's therapies.

Pr. Fei Lan Lab 2019.09 – 2021.07

Undergraduate Researcher | Focus: DNA Methylation & Epigenetics

Conducted fundamental wet-lab experiments focused on DNA methylation and epigenetic regulation mechanisms. Built a solid methodological foundation in molecular biology, cell culture, and standard biochemical assays.

Clinical Elective, HKU Queen Mary Hospital 2025.04 – 2025.05

Clinical Observer | Department of Medicine

Gained immersive clinical exposure through rounds and multidisciplinary case discussions. Observed the management of complex medical conditions in a premier academic medical center, bridging clinical practice with academic medicine.

EXTRACURRICULAR ACTIVITIES

- Volunteer Teaching & Cultural Investigation (Guizhou Zheng'an & Guangdong Puning)
- Fudan University Middle and Long Distance Running Team (Male 800m Bronze Medal)
- First Prize in China Biology Olympiads (Rank 9 in Guangdong Province)